

# TOI-4087b

R-2082

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_TOI4087\_250407 · created 2026-08-10T00:12:31+00:00

## BENCHMARK: MARGINAL

### Results

|                                                   |                                |
|---------------------------------------------------|--------------------------------|
| Mid-Transit Time (Tmid)                           | 2460772.934 +/- 0.0021 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.1342 +/- 0.0065              |
| Transit depth (Rp/Rs)^2                           | 1.8 +/- 0.18 [%]               |
| Orbital Inclination (inc)                         | 88.25 +/- 0.81                 |
| Airmass coefficient 1 (a1)                        | 1.08 +/- 0.01                  |
| Airmass coefficient 2 (a2)                        | -0.045 +/- 0.036               |
| Scatter in the residuals of the lightcurve fit is | 0.93 %                         |
| Best Comparison Star                              | #1 - [489, 384]                |
| Optimal Aperture                                  | 2.99                           |
| Optimal Annulus                                   | 7.64                           |
| Transit Duration (day)                            | 0.118 +/- 0.0029               |

### Accuracy vs published ephemeris (ExoClock/NEA)

|                                      |                                                                       |
|--------------------------------------|-----------------------------------------------------------------------|
| Rp/R $\blacksquare$ fit vs published | 0.1342 $\pm$ 0.0065 vs 0.10754068997360952 (deviation 2.18 $\sigma$ ) |
| Duration fit vs published            | 0.118 d vs 0.1242 d (deviation 1.13 $\sigma$ )                        |
| Mid-time O–C                         | -0.8 min                                                              |
| Depth SNR                            | 11.0                                                                  |
| Residual scatter                     | 0.93 %                                                                |

### Light curve

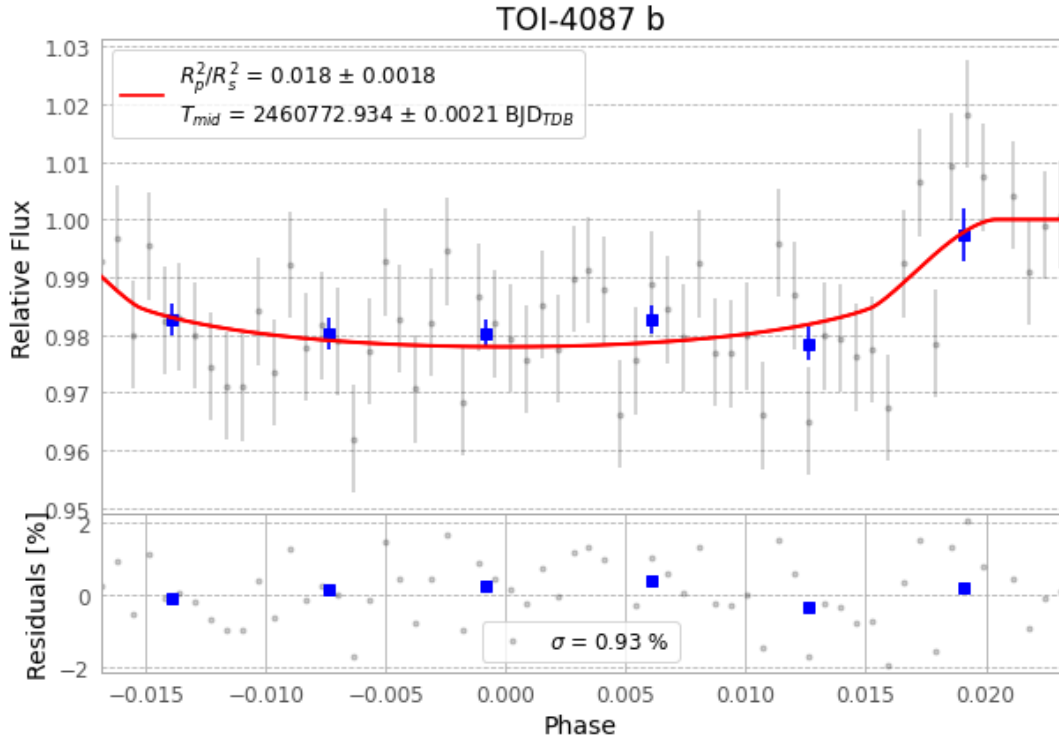


Figure 1 — FinalLightCurve\_TOI-4087 b\_2025-04-07.png (SHA-256 6ab5fe79934123d5...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [417, 285], 3 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 0821c2108b8c05a1...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit fd74a91

## Data provenance

62 FITS frames (41 MB), TOI4087250407090615.FITS → TOI4087250407120915.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TOI4087-250407.log (60e8dd009e07...), FinalLightCurve\_TOI-4087 b\_2025-04-07.png (6ab5fe799341...), FinalLightCurve\_TOI-4087 b\_2025-04-07.csv (8641b8881015...)

## Compute resources

Wall time 157.0 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

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Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-10 00:12Z.